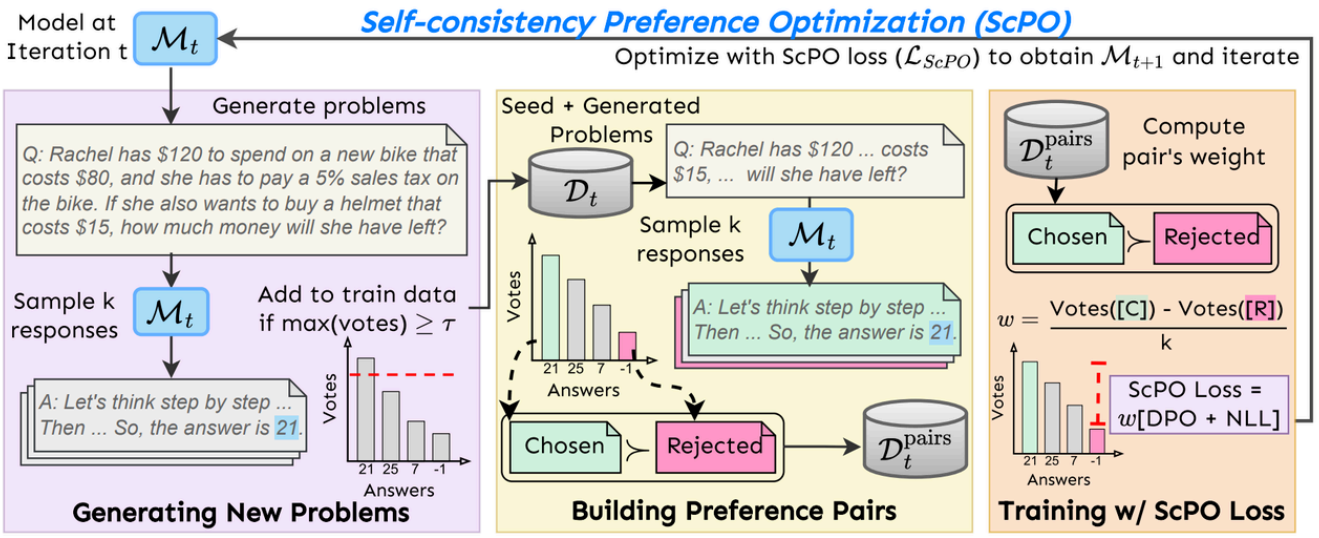
Self-Consistency Preference Optimization

> #DPO #Finetuning



- What problems does this framework solve?
  - The normal DPO is based on the construction of +- pairs. The preference of the answers needs to come from humans. Human labels are expensive and time costly.
  - Even with a reward model, when the problem is out of distribution, the reward can be not accurate.
  - A good way to achieve self-evaluation-based finetuning is by using self-consistency during the evaluation (introduce majority-voting to the finetune).
- What does this framework do?
  - **Data preparation**
    - Given a set of seed prompts and a base model M_0, we let the model generate a set of prompts based on the seeds.
    - To filter the generated prompts, we sample a set of responses from M_0, then count the # of each response. If none of the answers generated for this question receive a vote count exceeding the set threshold τ, then we discard this prompt.
  - **Preference Pair Construction**
    - Let the current model generate a set of answers (k answers) to each prompt **with temperature**. Then count the number of each answer (frequency). Let the answer with the highest frequency (*must be greater than τ, otherwise this prompt will be discarded*) be the positive and the one with the lowest frequency be the negative.
    - **Weighted Design**: If one answer has a significantly higher frequency, meaning that the model is more confident about that answer. The greater the difference of difference between the most-frequent answer and the least-frequent answer is, the higher quality the pair will be. Thus, let the weight be:

      $w(x) = (\mathcal{V}(y^+) - \mathcal{V}(y^-))/k.$
    - Objective:

$$\mathcal{L}_{SCPO}(y^+, y^-|x) = -w(x)\log\sigma\left(\beta\log\frac{M_\theta(y^+|x)}{M_t(y^+|x)} - \beta\log\frac{M_\theta(y^-|x)}{M_t(y^-|x)}\right) - w(x)\log M_\theta(y^+|x)$$

        - Weighted normal DPO+NLL lost: not only maximize the difference between the +- pairs, but also let the model learn the absolute distribution of the preferred answer y^+.
        - M_θ is the current model, and M_t is the previous model for reference.
    - Can be trained iteratively. Can naturally fit to the dataset with some of the data labeled: let the correct answer be y^+ and let the weight be 1 (maximum, most hard).
- Experiment Results
    - Pure unsupervised performance excels: On mathematical datasets like GSM8K and MATH, SCPO—operating entirely without ground truth labels—significantly outperforms both baseline models and methods using external reward models. It even approaches (with a gap < 1%) supervised training using ground truth labels ($IRPO_{Gold}$).
    - Semi-supervised surpasses supervised upper bounds: Combining ground truth labels with new questions self-generated by SCPO (semi-supervised mode) yields results that even outperform purely strong supervised baseline models.
    - Surpassing Complex Logical Reasoning: On the extremely challenging ZebraLogic logic puzzle task with no public training data, the SCPO-trained Llama-3 8B model outperformed much larger models like Llama-3 70B, Gemma-2 27B, and Claude-3 Haiku.
    - The necessity of weighted loss: Ablation experiments demonstrate that accuracy significantly declines without using vote-difference-based weighting ($w(x) = 1$), validating the effectiveness of integrating model confidence into the loss function.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM